

Academic Year 2026-27

Big Data Analytics (303105361)

Assignment 3

Chapter 3: Basics of Hadoop

5 Marks Questions

1. Explain Hadoop architecture with its components.
2. Compare RDBMS and Hadoop.
3. Explain the Hadoop Distributed File System (HDFS) architecture.
4. Describe the workflow of Map Reduce in Hadoop.
5. Explain shuffle and sort phases in Map Reduce.

4 Marks Questions

1. What is Hadoop? Explain its key features.
2. Explain the history and need of Hadoop.
3. Describe the components of Hadoop ecosystem.
4. Explain input formats and output formats in Map Reduce.
5. Explain task execution in Map Reduce.

2 Marks Questions

1. Define Hadoop.
2. Expand HDFS.
3. What is NameNode?
4. What is DataNode?
5. Define Map Reduce.
6. What is shuffle in Map Reduce?
7. Define input format.
8. Define output format.
9. State any two advantages of Hadoop.
10. What is task execution in Hadoop?